

A2 · Answer Key and Demo Script

CS425 · Stop Typing Your Password · instructor led

INSTRUCTOR COPY

This one is a follow along, not a group activity: you do each step on the projector, say "hands up when", and wait. Every transcript below is real output captured from a laptop and from `onyx.boisestate.edu` while this was written. Numbers will differ; the shapes will not.

Say this at the start

"Last week you typed your password a dozen times. By the end of today you will never type it again, and you will have a command called `probe` that you will use for the next two activities. Follow along; I will wait at every step." Then hand out the sheet and start step 1 on the projector.

BUDGET

STEPS	MINUTES	WHAT TO WAIT FOR
0, password login	10	Every hand up for a prompt on Onyx. One student may take the whole period; do not let the room wait on them.
1 to 2, key and copy	15	Every hand up for <code>ssh user@onyx uname -n</code> with no password. This is where Windows eats time.
3 to 4, config and timing	10	Four rows of timings on every sheet.
5 to 7, <code>bashrc</code> and <code>probe</code>	15	<code>ssh onyx probe example.com 80</code> working from a laptop.
8, alias	5	Q3 answered.
9, laptop	rest	Optional; whatever time is left.

STEP 0 · CAN THEY GET THERE AT ALL

MESSAGE	WHERE IT STOPPED	FIX
command not found / not recognized	The laptop; no client on the path	Git Bash. Do not install anything in class.
Could not resolve hostname	Name resolution	Spelling, then <code>curl -I https://www.boisestate.edu/</code> to prove the Wi-Fi works.
Connection timed out	The network; nothing answered	Phone hotspot. If that works, the first network filters 22.
Connection refused	A host, but not Onyx	Wrong hostname, nearly always.
Permission denied (publickey,password)	Onyx; the account	Username, not email. Slow retype. Three strikes: probably no Onyx account, note the name, move on.
REMOTE HOST IDENTIFICATION HAS CHANGED	Stale fingerprint on the laptop	<code>ssh-keygen -R onyx.boisestate.edu</code> , then yes.

The universal answer: `ssh -v user@onyx.boisestate.edu` and read the last lines before it stopped. Connecting to with nothing after it is the network; Authentications that can continue then a failure is the account. Put that on the board and leave it there; it is the one thing from today that transfers to every other host.

STEPS 1 TO 4 · KEYS AND MULTIPLEXING

Cold versus warm, measured from a laptop

```
$ ssh -O exit onyx ; time ssh onyx true
cold (full handshake): 1105 ms
```

```
$ time ssh onyx true
warm (reused master): 248 ms
```

```
$ ping -c 3 onyx.boisestate.edu
rtt min/avg/max = 31.4/33.7/35.3 ms
```

SAY: about 1.1 seconds against a 31 ms round trip. Ask what the extra time bought: a TCP handshake, a version exchange, a key exchange, and authentication. Accept any estimate in the range of a handful to a few dozen round trips. **DO NOT LET THEM CONCLUDE IT IS ALL NETWORK TIME:** key exchange is arithmetic, and on a fast link the crypto dominates. The warm case is roughly 4x faster and still not zero, because starting a process on the far end is not free.

Q1. Any estimate from about 5 to about 30 round trips passes, with something named: handshake, version exchange, key exchange, authentication. The largest speedup is normally the laptop with the slowest link or the slowest CPU; either explanation is fine if it matches their numbers.

Where step 2 goes wrong

- **No ssh-copy-id on Windows.** Manual procedure is on the page; have it on the board before you start step 2, not after the first hand goes up.
- **Still asks for a password.** Permissions, nearly always. On Onyx: `chmod 700 ~/.ssh; chmod 600 ~/.ssh/authorized_keys`. On the laptop: `chmod 600 ~/.ssh/config ~/.ssh/id_ed25519`. Run `ssh -v onyx` and read the line after "Offering public key".
- **Windows and ControlMaster.** Multiplexing is unreliable on Windows builds of OpenSSH. A Git Bash student may see a warning or two cold timings. Write both down, note the OS, move on.

STEPS 5 TO 8 · THE FUNCTION

Loaded and proven

```
$ cat ~/.bash_profile
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi
                                     ← the login shell sources .bashrc

$ head -20 ~/.bashrc
(no [ -z "$PS1" ] line on Onyx)

$ type probe
probe is a function

$ probe example.com 80
connect=0.033370 total=0.057519 code=200

$ ssh onyx probe example.com 80      # non-interactive, from the laptop
connect=0.033370 total=0.057519 code=200
```

SAY: the last one is the one that matters. That shell had no prompt and no terminal, and the function was still there, because bash reads `~/.bashrc` when a remote shell daemon starts it and Onyx has no guard line stopping it. **VERIFIED ON ONYX**, so if a student's fails, the cause is in their file, not in the machine. Usual causes: the edit did not save, or it went into a different file.

Q2. A guard line returns from `~/.bashrc` before the function is defined whenever the shell is not interactive, and `ssh host` command is not interactive, so the function never exists in that shell. Any sentence with "not interactive" and "returns before" or "never reaches" passes.

Q3. An alias is textual substitution at the front of a command, so the words you type can only land at the end. There is no way to put the second word into the middle of a URL. A function has positional parameters, `$1` and `$2`, and that is the whole reason `probe` is a function.

Rescue

- A **broken** `~/ .bashrc` can cost a student their shell. `ssh onyx -t 'bash --norc'` gives a clean shell to fix it in, unless the file exits or hangs.
- **vi**. On the board before step 3 and left there: `i` to type, `Esc` to stop, `:wq` save and quit, `:q!` bail out, `G` end of file, `o` new line below. Typed before `i` and things went strange: `Esc`, `:q!`, start over.
- **Missed the class**. The page is written to be done alone; point them at it and at A3's note that **probe** can be pasted at the prompt for one session.

PASS BAR: a row per member with two timings, and something written under each of the three questions. Award 20 or nothing.